A13485W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B6: CONDENSED-MATTER PHYSICS

TRINITY TERM 2021

Thursday, 10 June

Opening time: 09.30 am UK Time

You have two hours writing time to complete the paper and up to 30 minutes technical time to upload your answer file.

Answer **two** *questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspera during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) Simple X-ray sources are cheap and can be found in many laboratories. Neutron sources are very expensive. Give an example of when one might want to use neutron scattering rather than X-ray scattering in the study of a crystalline material. [2]

(b) A powdered sample of cubic $RbMnF_3$ is examined with neutron scattering at room temperature. The incoming neutrons have kinetic energy 25 meV. Measurements are made between total deflection angles $0°$ and $52°$. Diffraction rings are observed at total deflection angles $24.6°$, $35.1°$, $43.4°$ and $50.5°$. Determine the corresponding spacings of the lattice planes which cause these diffraction peaks. [6]

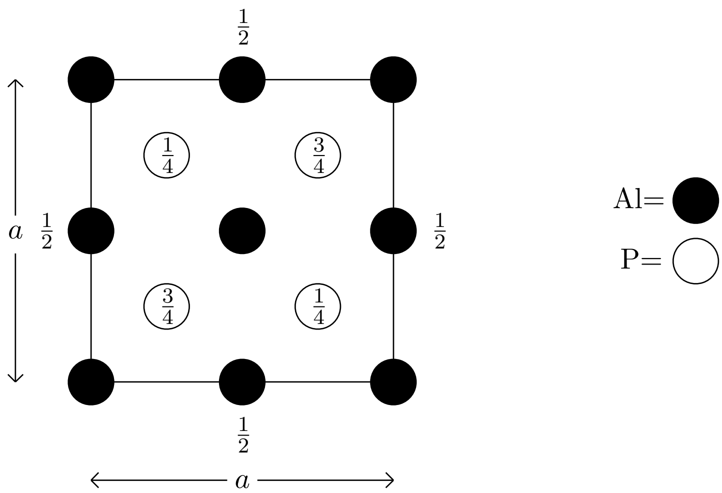
(c) Given that the lattice constant is known to be close to 0.4 nm, determine the Miller indices of these diffraction rings, deduce the lattice type, and find the lattice constant accurately. [8]

(d) When a magnetic field is applied to the sample, the diffraction rings are observed to change in intensity (but not in angle) with magnetic field. Suggest a reason for this observation. [3]

(e) The sample is now cooled down. At temperatures $T < T_c = 83\,\mathrm{K}$, two additional diffraction rings are observed (the rings which were present at room temperature do not measurably change their positions). One of the new rings has total deflection angle $21.3°$. Propose a physical mechanism for the appearance of these new rings. Predict the total deflection angle of the other new ring. [6]

The rest mass of a neutron is $m_n = 1.67493 \times 10^{-27}$ kg.

2. The following picture presents a plan view of the crystal aluminium phosphide (AlP) looking down the z-axis. The numbers attached to some atoms represent the heights of the atoms above the $z = 0$ plane expressed as a fraction of the cubic conventional unit cell edge a. Unlabelled atoms are at $z = 0$ and $z = a$.



(a) Determine the Bravais lattice type, list the coordinates of the lattice points in the cubic unit cell, and describe the basis. [6]

(b) Given an arbitrary direct lattice, give a general formula that defines the points of the reciprocal lattice. For aluminium phosphide, derive the reciprocal lattice (you must give a derivation, not just state the answer), determine the Bravais lattice type for the reciprocal lattice, and determine the lattice constant for the conventional unit cell of the reciprocal lattice. [7]

(c) How many (i) acoustic and (ii) optical phonon modes do you expect at a given wavevector in aluminium phosphide? Consider only acoustic waves in the $[1, 0, 0]$ direction. How many of these modes do you expect to be longitudinal and how many do you expect to be transverse? Explain how many *distinct* acoustic velocities do you would expect in this direction. [4]

(d) In the above diagram if we replace each Al atom with a silicon atom and also replace each P atom with a silicon atom, we obtain the crystal structure of silicon. The lattice constant of silicon is 0.5431 nm. The longitudinal acoustic velocity of silicon in the $[1, 0, 0]$ direction is $8433\,\mathrm{ms}^{-1}$. Estimate the longitudinal acoustic phonon energy at wavevector $\mathbf{k} = (2\pi/a, 0, 0)$. Express your answer in eV. Explain any approximations or assumptions you make. [3]

(e) Aluminium phosphide and silicon have very similar phonon spectra. However, there are also some minor differences. One difference occurs at the wavevector $\mathbf{k} = (2\pi/a, 0, 0)$. Explain how the spectra of aluminium phosophide and silicon differ at this wavevector and explain why this difference occurs. [5]

3. (a) Consider a two-dimensional insulating solid made of N_a atoms where the atoms in the solid only move in the two-dimensional plane. What would you predict for the heat capacity per atom C_v/N_a in the high temperature (classical) limit? Would C_v/N_a increase or decrease at lower temperature? Briefly justify both of your answers. [4]

(b) Consider a two-dimensional gas of N_e electrons of mass m. In the limit where the temperature T is high compared to the Fermi energy, $k_\mathrm{B}T \gg E_\mathrm{F}$, what is the heat capacity per electron C_v/N_e? Do you expect C_v/N_e to increase or decrease at lower temperature? Briefly justify your answers. [4]

For parts (c) and (d) consider a two-dimensional undoped semiconductor with a very small band gap E_gap. You may assume that the mass m_e^* of the electrons in the conduction band is equal to the mass of holes m_h^* in the valence band. We will assume that the band gap is small enough that $E_\mathrm{gap} \ll k_\mathrm{B}T$, so that we can set $E_\mathrm{gap} = 0$ in all calculations.

(c) Calculate the density of states as a function of energy. [3]

(d) Calculate the number N_c of electrons in the conduction band as a function of temperature. Let the total electronic heat capacity (including valence and conduction band) be called C_v^tot. Calculate $C_v^\mathrm{tot}/N_\mathrm{c}$. Your result should be greater than the quantities C_v/N_a and C_v/N_e calculated in parts (a) and (b) above in the high temperature limit. Briefly explain why your result for $C_v^\mathrm{tot}/N_\mathrm{c}$ does not violate the equipartition theorem. [11]

(e) If we choose a temperature such that in part (b) we have $k_\mathrm{B}T \ll E_\mathrm{F}$ but for parts (c) and (d) we have $k_\mathrm{B}T \gg E_\mathrm{gap}$, how does C_v^tot from part (d) compare to C_v from part (b)? Compute the ratio of these two quantities. [3]

The following integrals may be useful:

$$\int_0^\infty \mathrm{d}x \, \frac{1}{\mathrm{e}^x + 1} = \log(2) \approx 0.693 \qquad\qquad \int_0^\infty \mathrm{d}x \, \frac{x}{\mathrm{e}^x + 1} = \frac{\pi^2}{12} \approx 0.822$$

4. (a) Gallium arsenide (GaAs) is a covalently bonded direct band gap semiconductor with no localised magnetic moments on either atom (neglecting nuclear moments). Detailed data about GaAs are given below. Explain whether you would expect that undoped GaAs at zero temperature is a paramagnet, a diamagnet, or a ferromagnet. [2]

(b) The Larmor–Langevin magnetic susceptibility of a material is given by the formula

$$\chi = -\mu_0 \frac{\rho e^2}{6 m_\mathrm{e}} \langle r^2 \rangle$$

where ρ is the electron density and $\langle r^2 \rangle$ is the mean square radius of the atomic wavefunction. Determine the Larmor–Langevin susceptibility of undoped GaAs at zero temperature. Since the Larmor–Langevin calculation for a solid is rarely accurate to better than 30%, you may make rough approximations at the 30% level as long as you state your assumptions. [5]

(c) The sample is now very lightly doped with silicon which is an n-type dopant. At low temperature, the conduction band electrons bind to the dopant nuclei. Calculate the binding radius a_binding and the binding energy E_binding. Consider the magnetic susceptibility associated only with the electrons which are added by the dopants. There are two contributions to this susceptibility corresponding to the effect of the electron spin and the effect of the orbital motion. At what temperature do these two contributions cancel? [9]

(d) When the density of dopants is below $1/a^3_\mathrm{binding}$ the dopants form an impurity band with a bandwidth W (the energy difference between the bottom and top of the band) that can be approximated as

$$W = E_\mathrm{binding}\ \exp(-\alpha n^{-1/3}/a_\mathrm{binding})$$

with α a constant of order one. Briefly explain why W has this exponential form. [2]

(e) For the case of zero temperature, write an approximate expression for the magnetic susceptibility associated with the spin of the electrons in the impurity band as a function of the dopant density. You may set $\alpha = 1$ and you may make rough order of magnitude estimates. At roughly what density does this expression exceed the Larmor–Langevin susceptibility for the undoped GaAs? At this density, up to roughly what temperature is this calculation valid? [7]

<u>Gallium arsenide (GaAs) data</u>

Atomic numbers: gallium = 31, arsenic = 33
Atomic weights: gallium = 69.7 u, arsenic = 74.9 u
Density = 5.9 g cm^{-3}
Band gap = 1.4 eV
Conduction band effective mass = 0.067 m_e (with m_e the bare mass of the electron)
Conduction band g-factor = -0.45
Relative dielectric constant $\epsilon_\mathrm{r} = 12.9$